Structural Organisation In Animals

* Unicellular Organisms: All basic functions (digestion, respiration, reproduction) are performed by a single cell.
* Multicellular Organisms: Basic functions are carried out by different groups of cells.
* Example: Hydra - simple multicellular organism with thousands of each type.

Human body - composed of billions of cells.

* Tissues:
  - > Groups of similar cells + intercellular substances
  - > Perform a specific function.
  - > Complex animals have only four basic types of tissues.

* Organs:
  - > Tissues organized in specific proportion and pattern
  - > Examples: stomach, Lungs, heart, Kidney

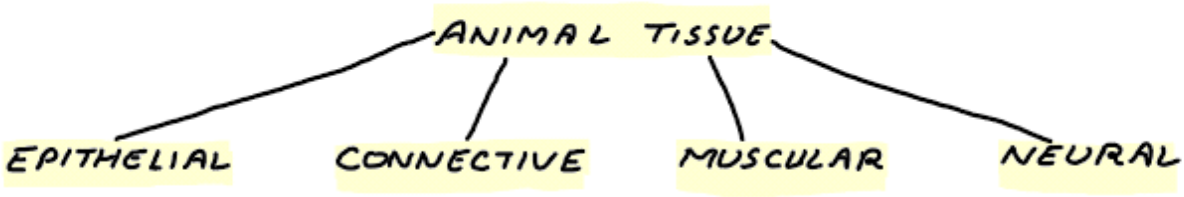
* Organ System:
  - > Two or more organs performing a common function (physical and/or chemical interaction).
  - > Examples: digestive system, respiratory system.

Division of Labour:

> Cells, tissues, organs, and organ systems divide up the work.

> Contributes to the survival of the body as a whole.

ANIMAL TISSUE :



* EPITHELIAL TISSUE :

> It has a free surface which faces body fluid or outside environment, hence forms covering.

> Cells are compactly packed with little intercellular space.

Epithelial Tissue

Simple epithelium

Basement membrane

Single layer, lining for body cavities, ducts & tubes.

Compound epithelium

2 or more layers of cells, protective function in skin. Cover dry surface of skin, moist surface of buccal cavity, pharynx, inner lining of salivary gland & pancreatic ducts.

Squamous epithelium

Flattened cells with irregular boundaries. Form diffusion boundaries. Eg- wall of blood vessel, air sacs of lungs.

Simple Squamous

Cuboidal epithelium

Cube like cells. Perform secretion & absorption. Eg- ducts of glands, tubular part of nephron. Brush border cuboidal cells are present in PCT.

Simple cuboidal

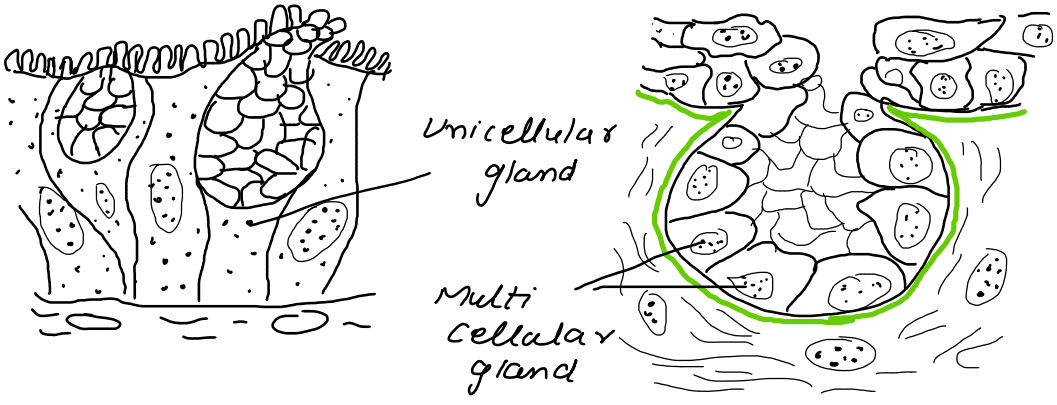
Columnar epithelium

Tall, slender cells with nuclei at the base. Help in secretion & absorption Eg- Lining of stomach, intestine.

Simple columnar

* **Ciliated epithelium:** May be cuboidal/columnar, if cells bear cilia on their free surface. Function is to move particles or mucus in a specific direction. Present inside hollow organs like bronchioles, fallopian tube.

Glandular epithelium: May be UNICELLULAR- isolated glandular cells (eg - goblet cell) or MULTICELLULAR – cluster of cells (e - salivary gland). May be EXOCRINE - have duct (eg - mucus, saliva, ear wax, oil milk) or ENDOCRINE – ductless (eg - thyroid etc)



* Cell Junction :

< Tight junction - To stop substance to leak across a tissue.

< Adhering junction - Cement neighbouring cells together.

< Gap junction - Facilitate cell to cell communication by connecting the cytoplasm of adjacent cells, allowing rapid transfer of ions, small molecules, and sometimes big molecules.

Connective Tissue :

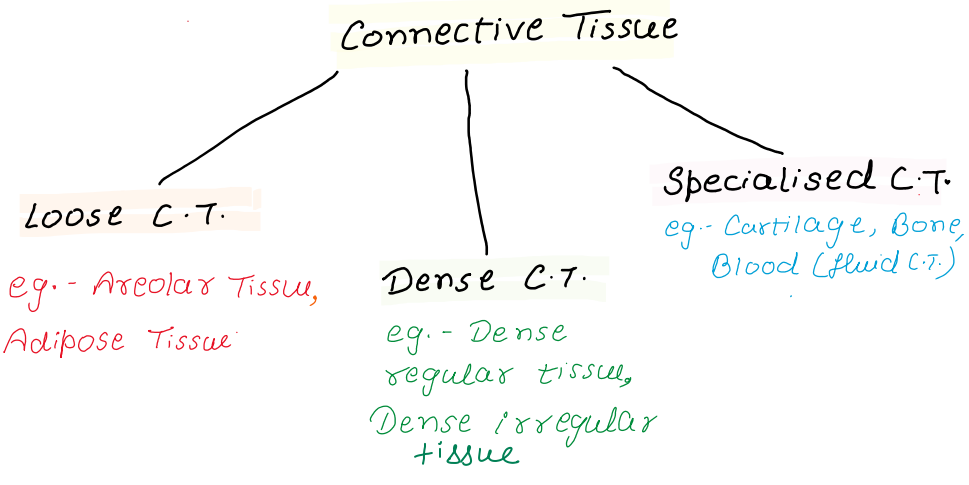
- Most abundant and widely distributed in complex animals.
- **Function** : linking and supporting other tissue/organs.
- Connective tissues except blood, cells secrete →

Cells secrete

↙ Fibres of structural Proteins (collagen, elastin)

↘ Modified polysaccharides (accumulate b/w cells and fibres act as matrix)

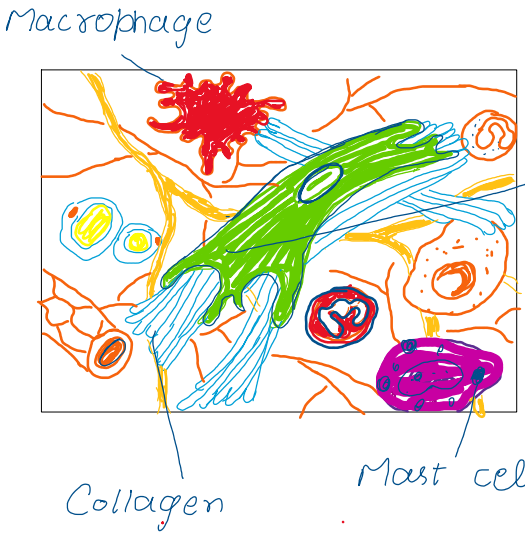
Types of C.T. -



Loose C.T.

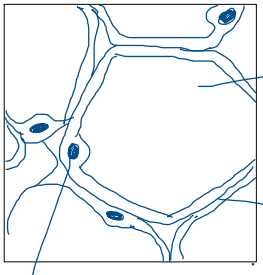
Areolar Tissue

present beneath the skin.
Serves as a support framework for epithelium.
Contains fibroblasts, macrophages & mast cells.



Adipose Tissue

- located beneath the skin.
- Cells are specialised to store fats.
- It stores excess (unused) nutrients as fat after conversion.



Dense C.T.

Dense regular T.

- Orientation of fibres show regular pattern
- Collagen fibres are present in rows b/w many parallel bundles of fibres.
- eg. - Tendons (attach skeletal muscles to Bones), Ligaments (attach one Bone to another)

Dense irregular Tissue

- Orientation of fibres show irregular pattern.
- Many fibres and fibroblasts are oriented differently.
- Present in - Skin

Specialised C.T.

Cartilage

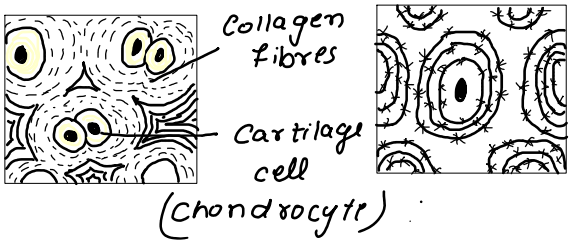
- Intercellular material is solid and pliable, resist compression.
- Chondrocytes - enclosed in small cavities within matrix.
- Most of the cartilage in vertebrate embryos are replaced by bones in adults.
- It is not in the tip of nose, outer ear joints, b/w adjacent bones of the vertebral column, limbs & hands in adults.

Bones

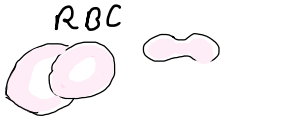
- Have a hard & non-pliable ground substance rich in Ca^{++} salts & collagen fibres.
- It supports softer tissues & organs.
- Osteocytes are not in the spaces called lacunae.
- Bone marrow in some bones is the site of production of blood cells.

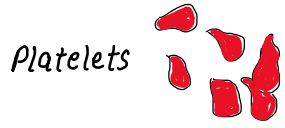
Blood

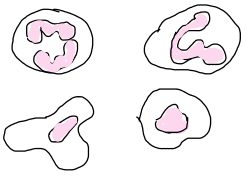
- It is a fluid connective tissue
- It contain plasma, RBC, WBC & platelets.
- It is a main circulating fluid, helps in transport of various substances.



(Chondrocyte)

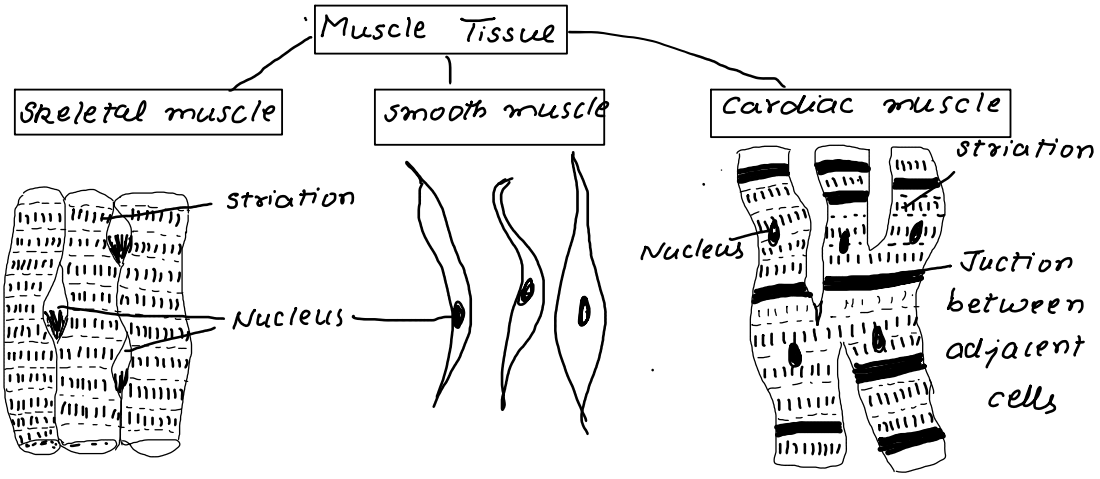






Muscle Tissue

- Made of many long, cylindrical fibres arranged in parallel arrays.
- Fibres - composed of numerous fine fibrils (myofibrils)
- Muscle fibres contract (shorten) in response to stimulation, then relax (lengthen)
- Muscles are of three types.



- Closely attached to skeletal bones.
- Striated, bundled together in parallel fashion.
- Voluntary in action.

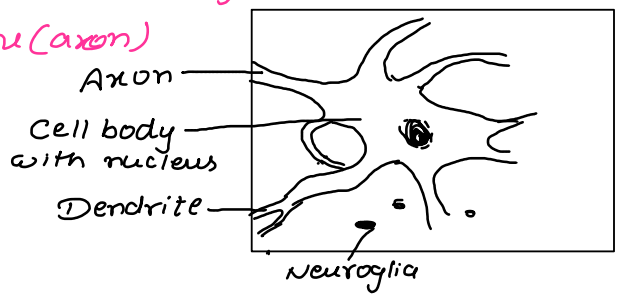
- Fibres taper at both ends (fusiform)
- No striation
- Bundled together in a connective tissue.
- Found in walls of internal organs (stomach, intestine, blood vessels)
- Involuntary

- Present only in Heart.
- Intercalated disc allow cell to contract as a unit.
- Cell junctions fuse cardiac muscle cell membranes, making cells stick together.
- Involuntary.

Neural Tissue :

- It exerts the greatest control over the body's responsiveness to changing conditions.
- Neurons are excitable cells.

- Neuroglial cells - make up more than one half the volume of neural tissue in our body.
- Neuron stimulation generates an electrical disturbance.
- Electrical disturbance swiftly travels along neuron plasma membrane (axon)



COCKROACH

General Features:

- Periplaneta americana are about 34-53 mm long with the wings that extend beyond the tip of the abdomen in males.

Phylum - Arthropoda

Class - Insecta

- Nocturnal - Active during night
- Omnivores

Morphology:

BODY (covered with hard chitinous Exoskeleton)
- Head - fusion of 6 segments
- Thorax - 3 distinct segments
- Abdomen - 10 segments

- Each segment is covered with chitinous exoskeleton (sclerites).

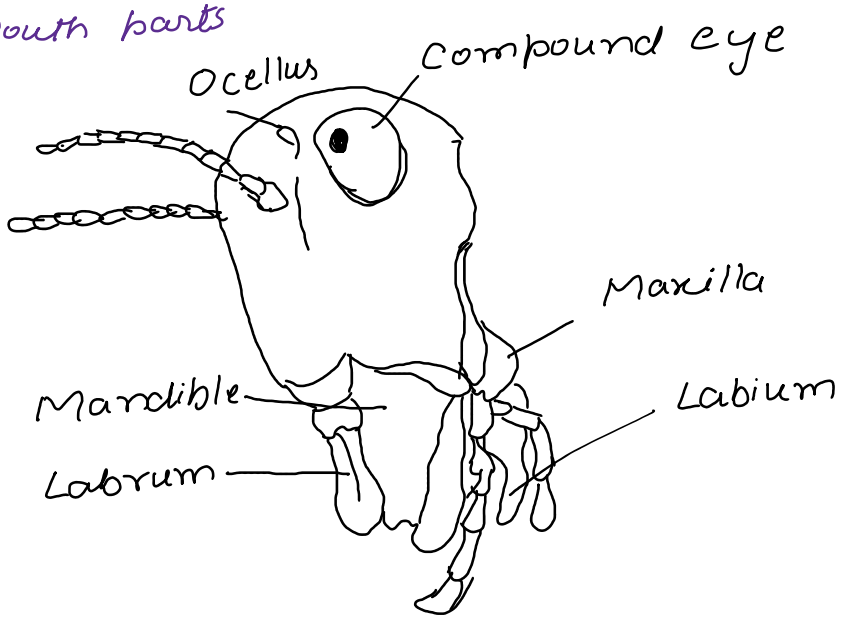
Dorsally - Tergites

Ventrally - sternite

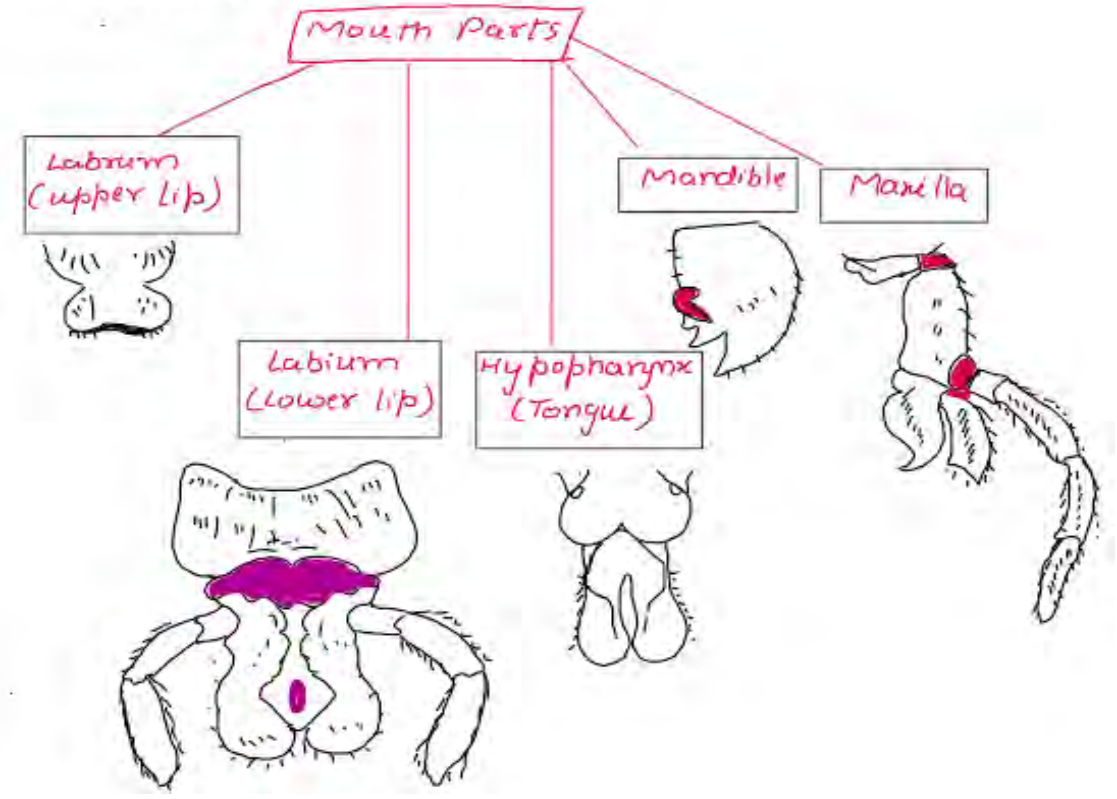
- Arthrodial membrane: flexible articular membrane connect the sclerites.

Head of Cockroach:

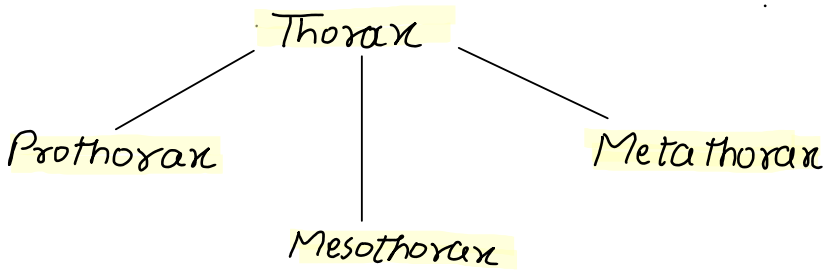
- Shape - Triangular
- Lies anteriorly at right angles to the longitudinal body axis.
- formed by fusion of 6 segments.
- Head is connected to thorax by short extension of Prothorax known as neck.
- Due to flexible neck head shows great mobility.
- Head Capsule bears:
  - Compound eyes
  - Antennae
  - Mouth parts



Mouth Parts:



Thorax



Wings -

Forewings / Mesothoracic
First pair of wings
- also known as Tegmina
- not used in flight
- cover the hind wings when at rest.
- Opaque, dark and leathery.

Hindwings / Metathoracic
second pair of wings
- Transparent, membranous
- Used in flight.

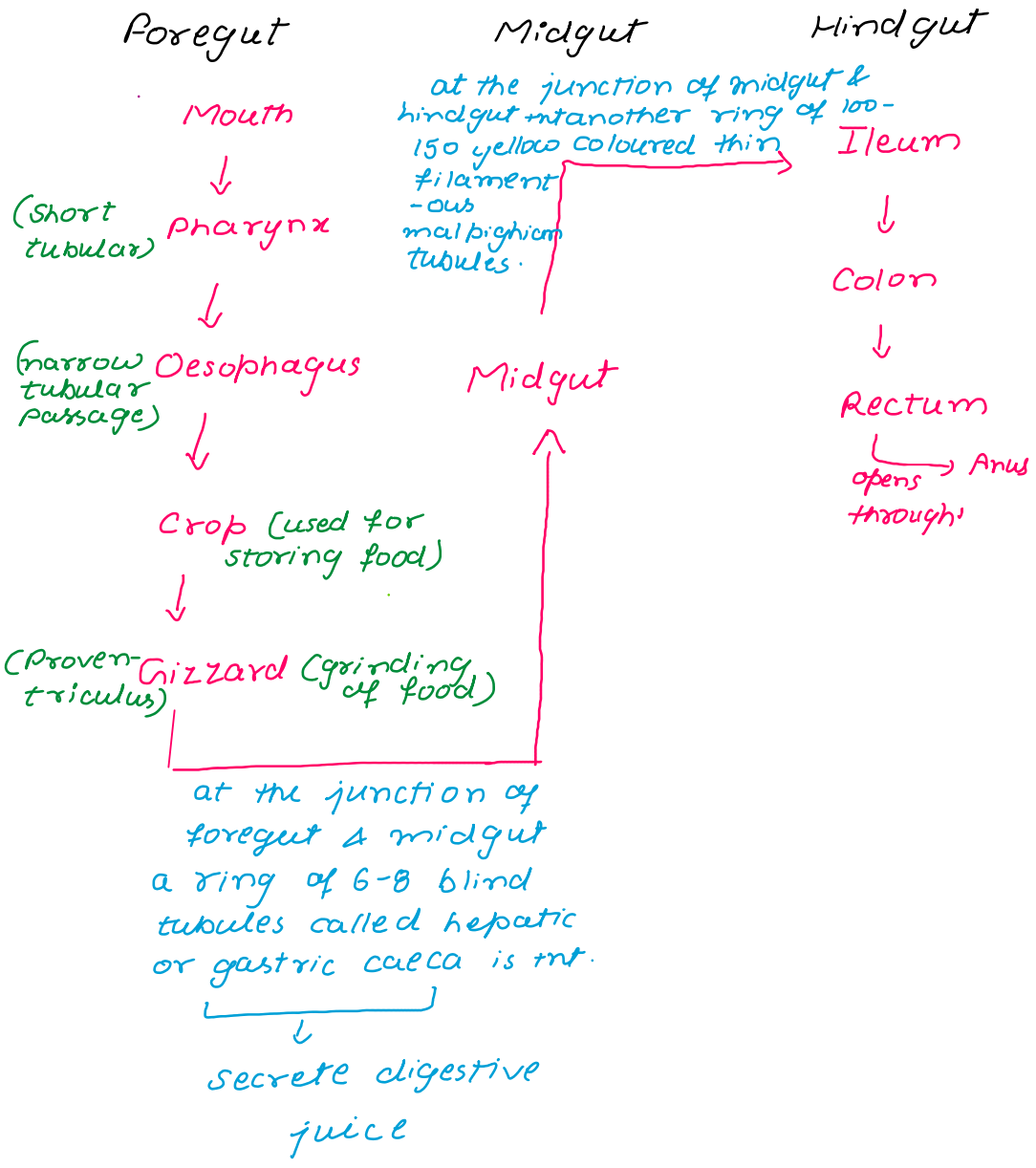
Abdomen

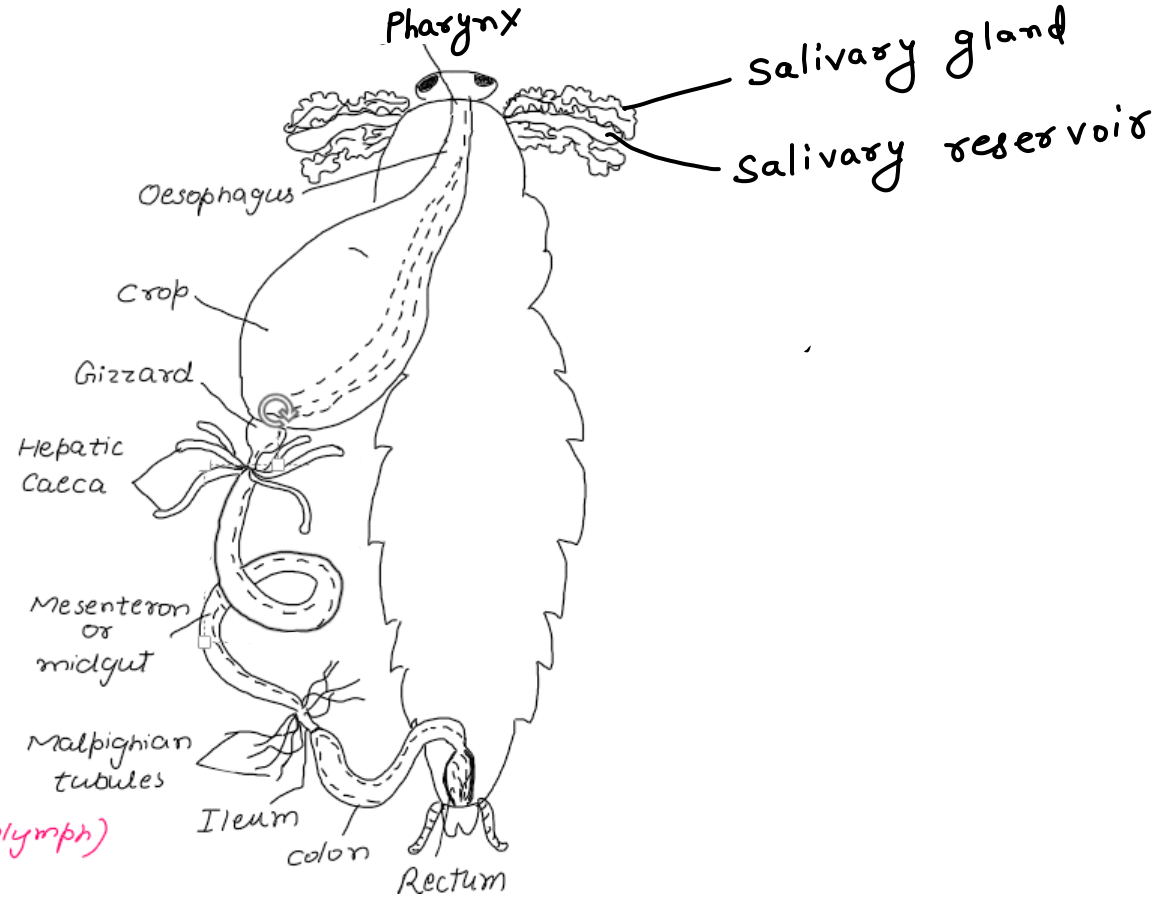
- In both males and females - 10 segments
- In females, brood or genital pouch: formed by fusion of 7th, 8th, 9th sternum.
  Boat shaped
- Anterior part of genital pouch contains female gonopore, spermathecal pores & Collateral glands.
- In males, genital pouch or chamber lies at the hind end of abdomen bounded dorsally by 9th and 10th terga & ventrally by 9th sternum.
- Male genital pouch contain - Dorsal anus, ventral male genital pore & gonapophysis.

ANATOMY

Digestive System

- Alimentary canal present in the body cavity is divided into three regions:



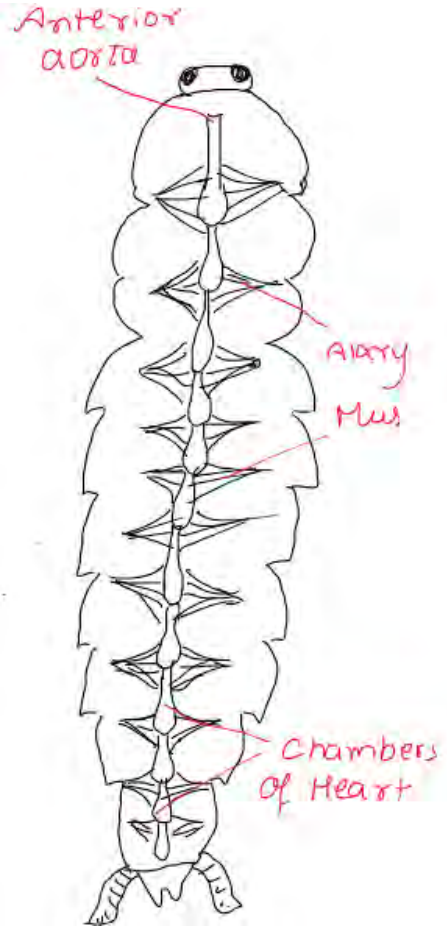


(Removal of excretory products from haemolymph)

Blood Vascular System

- Type: Open circulatory system
- Blood vessels: Poorly developed
- Haemocoel: Blood vessels open into a large body cavity called the haemocoel.
- Haemolymph:
  - Visceral organs are bathed in haemolymph within the haemocoel.
  - Composition: -Colourless plasma. (No RBCs)
    - Haemocytes (Blood cells)

- Heart:
  - Elongated muscular tube
  - Location: mid dorsal line of thorax & abdomen.
  - Structure: Differentiated into funnel-shaped chambers
  - Ostia: Each chamber has opening called ostia on either side.
  - 13 chambered and b/w two chambers fan shaped Alary muscles (12 pairs)



Blood flow

- Blood (Haemolymph) → Haemocoel/sinuses → Ostia → Heart → pumps haemolymph anteriorly → toward → Sinuses

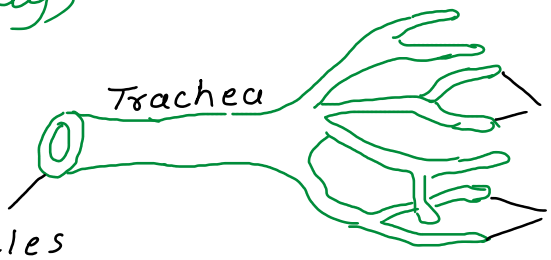
Respiratory System

- Consists of - Network of Trachea
- Trachea open through - 10 pairs of spiracles

- Air move

Spiracles $\xrightarrow{O_2}$ Trachea $\xrightarrow{O_2}$ Tracheoles $\xrightarrow{O_2}$ all parts of body

(+nt on the lateral side of the body)



Tracheoles (exchange of gases take place by diffusion)

Spiracles (regulated by sphincters)

Excretory System

- Excretion performed by Malpighian tubules.
- Structure of tubules: Lined by glandular and ciliated cells
- Function of Tubules:
  - Absorb nitrogenous waste from haemolymph
  - Convert nitrogenous wastes into uric acid.
- Cockroach is called Uricotelic.
- Fat body, nephrocytes & urecose glands also help in excretion.

Nervous System

- Consists of a series of fused, segmentally arranged ganglia

  3 ganglia – Thorax

6 ganglia – Abdomen

- Head holds a bit of a nervous system, rest situated along the ventral (belly-side) part of its body.
- Brain is represented by supra-oesophageal ganglion, supplies nerves to antennae and compound eyes.
- Sense organ are antennae, eyes, maxillary palp, labial palp, anal cerci etc.
- Compound eyes- situated at the dorsal surface of head.
- 2000 Hexagonal ommatidia – Each eye
- Mosaic vision (more sensitivity but less resolution) Nocturnal vision.

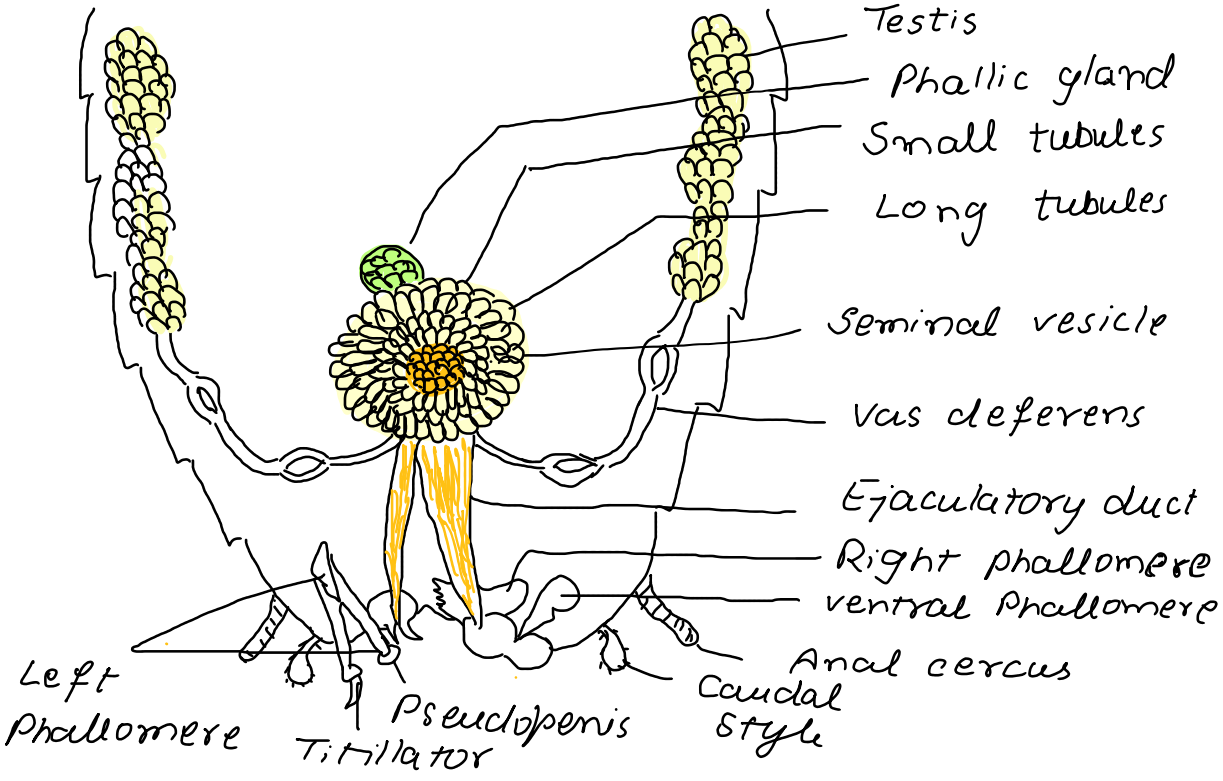
Reproductive System

Cockroach – Dioecious (sexes are separate)

Male Reproductive System

- Testes: A pair, lateral side of 4th–6th abdominal segments.
- Vas deferens: Thin tube from each testis.
- Seminal vesicle: stores sperms, connects vas deferens to ejaculatory duct.

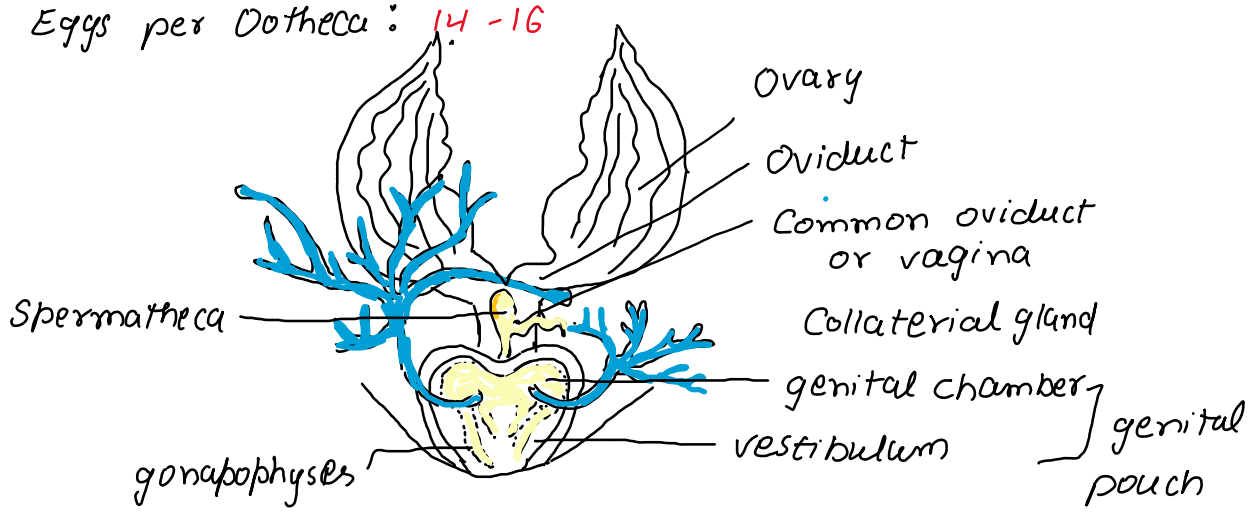
- Ejaculatory duct: Opens to male gonopore (ventral to anus)
- Mushroom-shaped gland: Accessory reproductive gland (6th-7th abdominal seg.)
- External genitalia: Male gonapophysis/ phallomere (chitinous, asymmetrical, surround gonopore).
- Sperms: Stored in seminal vesicles, glued in bundles called spermatophores, discharged during copulation.



Female Reproductive System

- Ovaries: Two large, lateral sides of 2nd-6th abdominal segments.

- Ovarian tubules (ovarioles): 8 in each ovary, contain developing ova (chain).
- Oviducts: Unite to form a single median oviduct (vagina).
- Genital chamber: Vagina opens into it.
- Spermatheca: A pair in the 6th seg., opens into the genital chamber (stores sperms).
- Fertilization: Sperms transferred via spermatophores.
- Oothecae: Fertilized eggs encased in dark reddish to blackish brown capsules (8 mm long)
- Oothecae Deposition: Dropped or glued to suitable surface (damp, near food).
- Oothecae Production: Average 9-10 per female.
- Eggs per Ootheca: 14 - 16



Development

- Paurometabolous: Development through nymphal stages.
- Nymphs: Resemble adults.
- Moulting: Nymphs moult 13 times to reach adulthood.
- Wing Pads: Next to last nymphal stage has wing pads.
- Wings: Only adults have wings.

Economic Importance:

- Most species are wild, no known economic importance.
- Few species thrive in human habitats.
- Pests:
  - Spoil food.
  - Contaminate food with smelly excreta.
  - Disease vectors: Transmit various bacterial diseases by contaminating food.

FROG

Kingdom: Animalia

Phylum: Chordata

Class : Amphibia

- Most common species of frog found in India
  - Rana tigrina
- They belong - AMPHIBIA
  dual life

- They can live both on land as well as in water.
- They are poikilotherms / cold blooded (don't have constant body temp.)
- During peak summer & winter they take shelter in deep burrows to protect them from extreme heat and cold.
- Summer sleep - Aestivation
  Winter sleep - Hibernation
- Camouflage - Ability to change colour to hide from enemies.
- Mimicry : Protective coloration for camouflage.

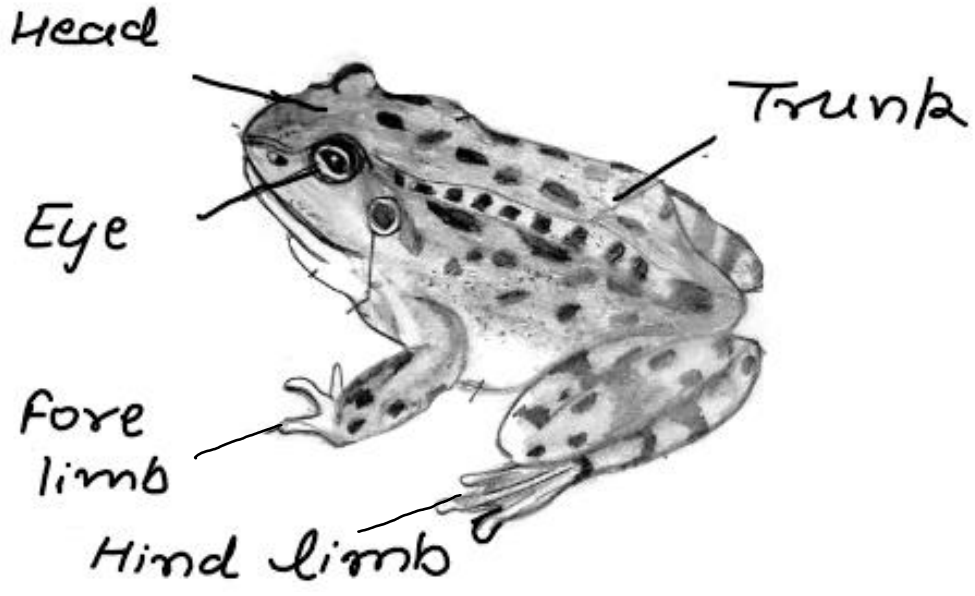
Morphology :

- Body is divisible into head and trunk only (neck and tail absent)
- Skin:
  - Smooth and slippery due to mucus.
  - Maintained moist.
  - Dorsal side : Olive green with dark irregular spots.
  - Ventral side : Uniformly pale yellow
  - water absorption : Absorbed through skin (does not drink)

- Head
  - Nostrils: A pair above the mouth.
  - Eyes: Bulged, covered by a nictitating memb. (protection in water)
  - Tympanum: Membranous ear on either side of eyes (receives sound)

- Limbs:
  - Forelimbs: shorter, end in 4 digits. Helps in walking, leaping, burrowing.
  - Hindlimbs: Longer and muscular, end in 5 webbed digits. Helps in swimming, leaping, burrowing.

- Sexual Dimorphism:
  - Males: Possess sound-producing vocal sacs.
  - Males: Possess a copulatory pad on the first digits of forelimbs (-nt in females)

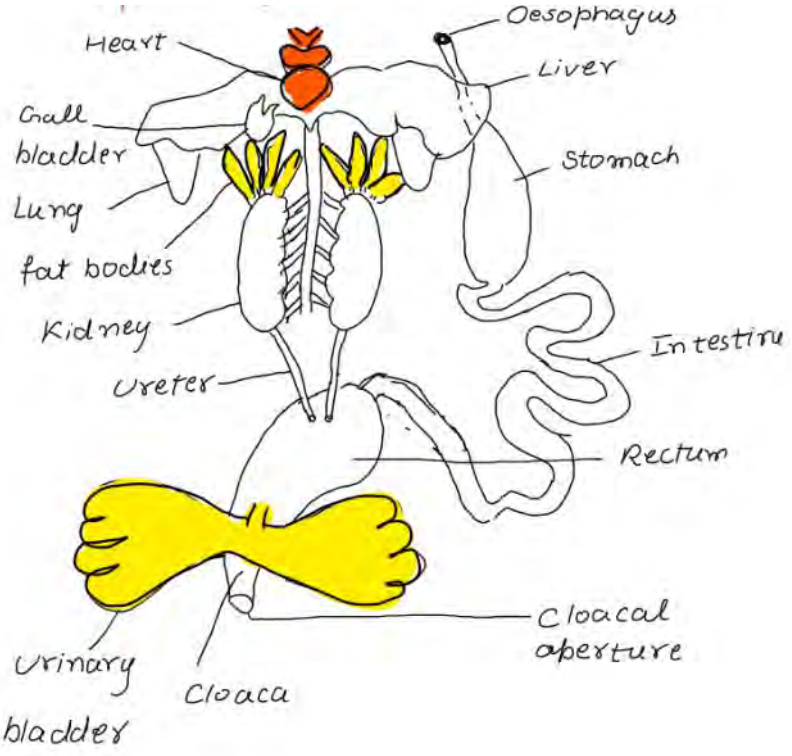


Digestive System:

- Alimentary canal + Digestive glands
  - ↓ Simple tube, short (carnivorous diet)
  - ↓ Liver, Pancreas

- Mouth → Buccal cavity → Pharynx → Oesophagus (short) → Stomach → Intestine (reduced length) → Rectum → Cloaca

- Tongue: Bilobed, for capturing food
- Stomach: Digestion by HCl and gastric juices (forms chyme)
- Small intestine (duodenum): Receives bile (gall bladder - emulsifies fats) and pancreatic juice (pancreas - digests carbs, proteins) via common bile duct
- Intestine: final digestion
- Absorption: Villi and microvilli in the inner wall of intestine.
- Undigested waste: Rectum → Cloaca



Respiratory System:

- Aquatic Respirⁿ (water): Cutaneous respiration (skin diffusion of dissolved O_2)
- Terrestrial Respirⁿ (Land):
  - Buccal cavity
  - skin
  - Lungs (Pulmonary respiration): paired, elongated, pink, sac-like in thorax.

    Nostrils → Buccal cavity → Lungs

- Aestivation & Hibernation: Cutaneous respiration

Circulatory System:

- Type: Well-developed closed type.
- Components → Heart, Blood, Blood vessels, Lymphatic system (lymph, lymph channels, lymph nodes)

- Heart :
  - Location : Upper body cavity.
  - Chambers : Three (2 atria, 1 ventricle)

  - Covering : Pericardium
  - Sinus venosus : Triangular, joins right atrium, receives blood from vena cava.
  - Conus Arteriosus : Sac like, ventral to ventricle, ventricle open into it.

- Arterial System : Carries blood from Heart to body
- Venous System : Collects blood from body to heart (vena cava)
- Portal Systems
  - Hepatic portal system : Liver and intestine connection.
  - Renal portal System : Kidney and lower body parts connection.
- Blood
  - Plasma
  - RBCs (nucleated, Hb)
  - WBCs (leucocytes)
  - Platelets

- Lymph : Lacks few proteins and RBCs.
- Function of blood : Transport nutrients, gases, water.
- Mechanism : Pumping action of the muscular Heart.

Excretory System:

- Components
  - Kidneys (compact, dark red, bean-like, posterior)
  - Ureters
    - Males - Urinogenital duct (Ureter + sperm duct) → Cloaca
    - Females - opens into cloaca
  - Cloaca
  - Urinary bladder (thin-walled, ventral to rectum)

- Kidney Units: Uriniferous tubules or nephrons.
- Ureters:
  - Males: Urinogenital duct (ureter + sperm duct) → Cloaca
  - Females: Ureters open separately into the cloaca.
- Urinary Bladder: Stores urine, opens into cloaca.
- Nitrogenous waste: Urea (ureotelic animal).
- Mechanism: Blood carries waste to kidneys for separation and excretion.

- Neural Control and Coordination:

- Neural System: CNS (brain & spinal cord), PNS (cranial & spinal nerves), ANS (Sympathetic and Parasympathetic)

- Cranial Nerves: Ten pairs.

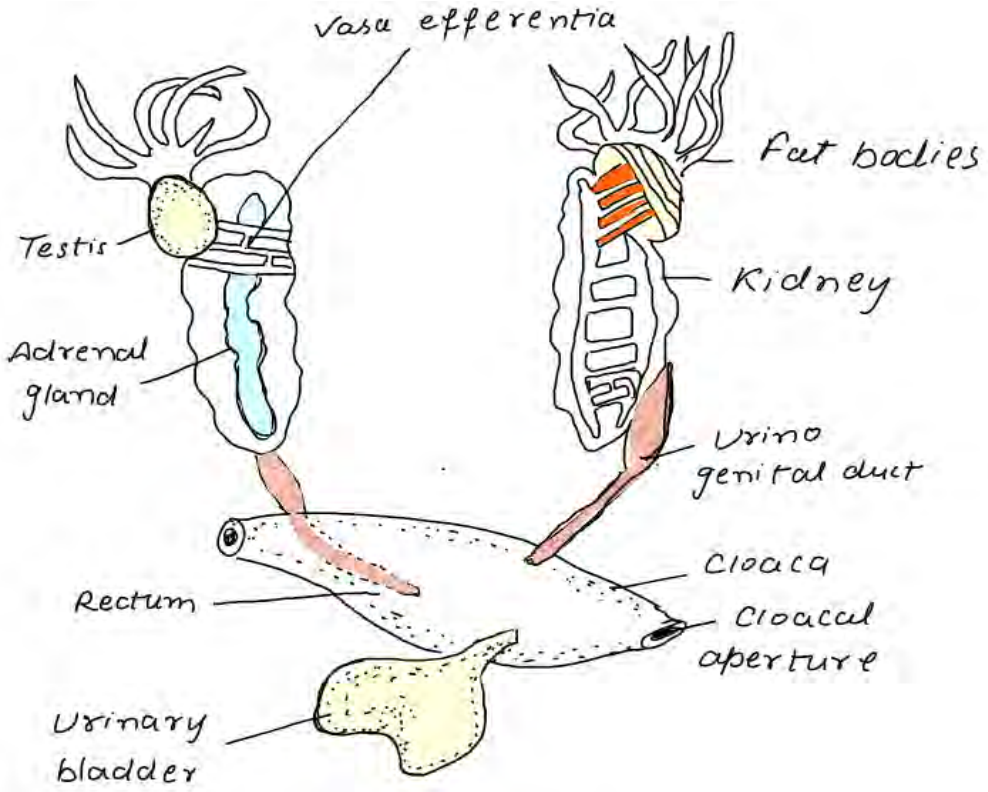
- Brain: Enclosed in Cranium (brain box)
  - Forebrain: Olfactory lobes, paired cerebral hemispheres, unpaired diencephalon.
  - Midbrain: Paired optic lobes.
  - Hindbrain: Cerebellum, medulla oblongata.
- Spinal Cord: Continuation of medulla oblongata through foramen magnum, enclosed in vertebral column.
- Endocrine Glands: Pituitary, thyroid, parathyroid, thymus, pineal body, pancreatic islets, adrenals, gonads (chemical coordination via hormones).

Sense Organs:

- Touch: Sensory papillae
- Taste: Taste buds
- Smell: Nasal epithelium
- Vision: Eyes (paired, spherical in orbits, simple - one unit).
- External ear absent, tympanum visible.
- Hearing & Balance: Tympanum with internal ears.

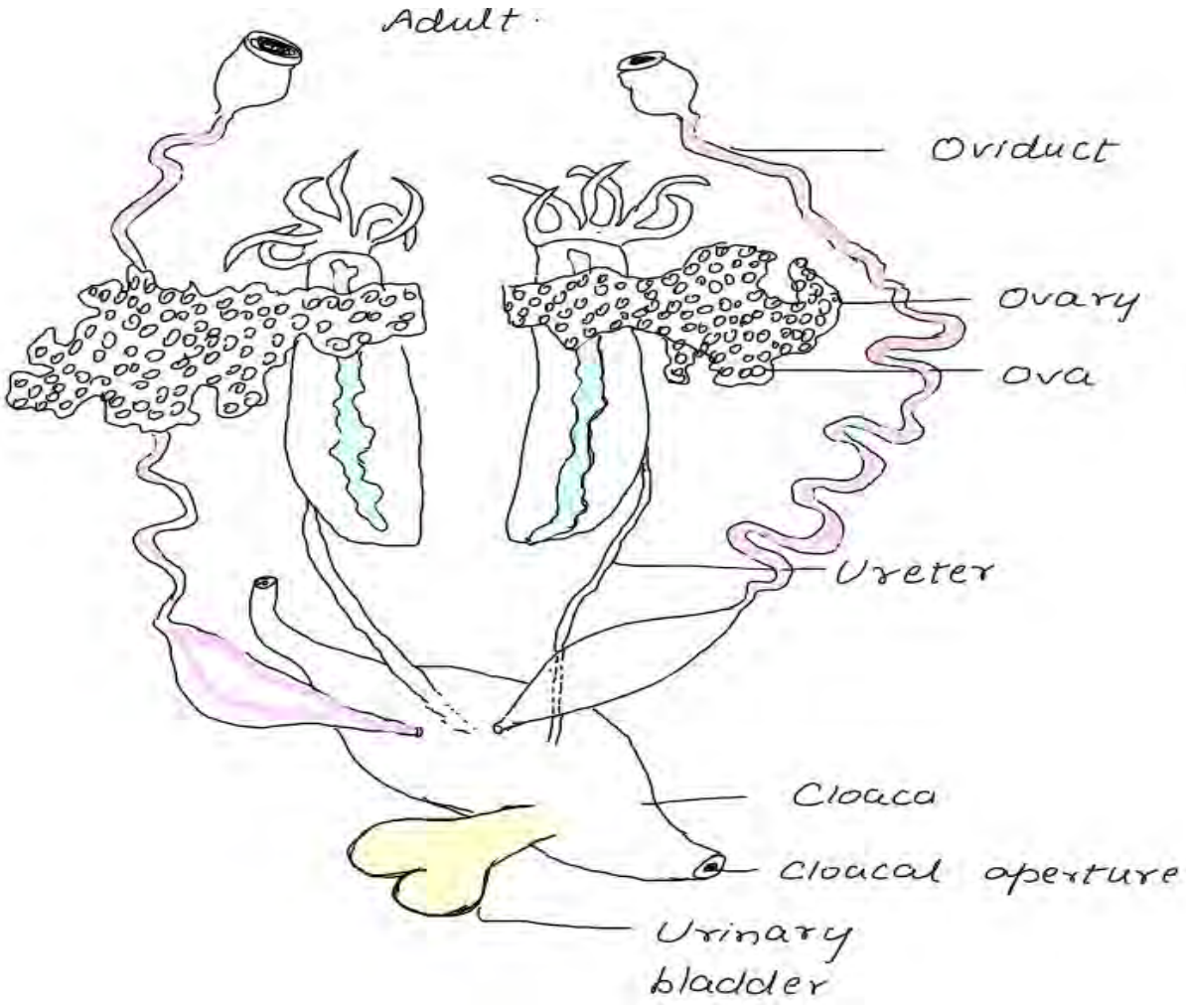
Reproductive System:

- Male:
  - Testes: Paired, yellowish ovoid, adhered to kidneys by mesorchium.
  - Vasa efferentia (10-12 in no. that arise from testes) ⟶ Kidneys ⟶ Bidder's canal ⟶ urinogenital duct ⟶ Cloaca (common passage for faecal matter, urine, sperms)



Male Reproductive System

- Female:
  - Ovaries: Paired, near kidneys (no functional connection)
  - Oviducts: Paired, from ovaries → open separately into cloaca.
  - Ova Production: 2500 - 3000 at a time
  - Fertilization: External, in water.
  - Development: Larval stage (tadpole) → Metamorphosis → Adult.



Ecological Importance:

- Eats insects (protects crops)
- Maintains ecological balance (food chain/ web link).
- Food source for humans in some countries (muscular legs).